01. Business Problem & Objective

Frame the prediction task and the decision it supports.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Business problem

CreditWise aims to predict whether a loan application will be approved using applicant financial, demographic, employment and loan attributes. A model could support preliminary risk triage and help analysts prioritize applications, but it must not replace regulated credit assessment or explainable human review.

ML framing

This is a supervised binary classification problem. The target is Loan_Approved, with two observed classes: Yes and No. The notebook uses LabelEncoder, which maps the two labels to numeric classes (alphabetically, No=0 and Yes=1).

- Estimate the probability of approval from information available at application time.
- Compare interpretable and non-linear baseline classifiers.
- Understand the precision/recall trade-off for the approved class.
- Identify data-quality and fairness issues before any operational use.

Business success criteria

Accuracy alone is insufficient. False approvals can create credit losses; false rejections can exclude creditworthy applicants and reduce business. The decision threshold should be chosen using the relative costs, policy constraints, calibration and fairness checks. The model should be evaluated on a representative, untouched holdout and monitored for drift.

Scope and actual repository workflow

- Dataset has 1,000 rows and 20 columns, with 50 missing values in every column, including Loan_Approved.
- Notebook imputes numeric columns with mean and object columns with most-frequent value before the split.
- It removes Applicant_ID, encodes Education_Level and target with LabelEncoder, one-hot encodes six other categorical predictors, and scales features.
- It splits 80:20 with random_state=42, but does not stratify.
- It compares Logistic Regression, KNN and Gaussian Naive Bayes; then adds squared DTI/Credit Score features and reruns the models.
- No deployment section is included.

Interview framing

> “The objective was to build a binary classifier that estimates loan approval from applicant and loan characteristics. I compared simple baselines and evaluated precision, recall, F1 and accuracy. I also identified an important data issue: the notebook imputes missing target labels before splitting, which can create artificial outcomes. I would correct that before trusting the reported performance.”